# Novel Prep Learning Center
## Math I / Enhanced Math I Placement Exam
IUSD-Aligned Readiness Diagnostic

Name: ____________________ Date: ____________________

**Directions:** Read each question carefully. Choose the best answer for multiple-choice questions. Show all work for graphing and free-response questions. Simplify all final answers.

**Teacher Use:** This diagnostic is designed to answer two placement questions. First, is the student ready for Math I? Second, if Math I readiness is secure, is the student ready for Enhanced Math I? Part A checks core algebra. Part B measures Math I outcomes. Part C and the extended free-response tasks emphasize Enhanced Math I readiness signals: radicals, Pythagorean reasoning, stronger modeling, systems, transformations, coordinate geometry, and proof.

| | |
|---|---|
| **Purpose** | Gate 1 confirms Math I readiness; Gate 2 confirms Enhanced Math I readiness after Gate 1 is secure. |
| **Structure** | 50 multiple-choice questions, graphing, free response, scoring guide, study plan, and answer key. |
| **Use** | Placement decisions should prioritize the two diagnostic gates. |

## Multiple Choice Questions

Questions 1-30 are used primarily for Math I readiness. Questions 31-50 add Enhanced Math I readiness signals.

### Part A — Core Readiness

1 Solve for $x$: $-3x + 7 = 25$.

(A) $x = -6$
(B) $x = 6$
(C) $x = -\frac{32}{3}$
(D) $x = \frac{32}{3}$

2 Which expression is equivalent to $4(2x - 5) - 3(x + 1)$?

(A) $5x - 23$
(B) $11x - 17$
(C) $5x - 17$
(D) $8x - 8$

3 Solve: $\frac{2x-5}{3} = 7$.

(A) $x = 8$
(B) $x = 11$
(C) $x = 13$
(D) $x = 26$

4 Which inequality represents the solution to $-4x + 9 < 21$?

(A) $x < -3$
(B) $x > -3$
(C) $x < 3$
(D) $x > 3$

5 Which relation is a function?

(A) $\{(1, 2), (1, 4), (3, 6)\}$
(B) $\{(-2, 5), (0, 5), (2, 5)\}$
(C) $\{(3, 1), (3, 2), (3, 3)\}$
(D) $\{(0, 0), (0, 1), (1, 0)\}$

6 The slope of the line through $(-2, 5)$ and $(4, -7)$ is:

(A) $2$
(B) $-2$
(C) $\frac{1}{2}$
(D) $-\frac{1}{2}$

7 What is the $y$-intercept of $3x - 2y = 10$?

(A) $(0, -5)$
(B) $(0, 5)$
(C) $(\frac{10}{3}, 0)$
(D) $(-5, 0)$

8 Which equation has infinitely many solutions?

(A) $2(x + 3) = 2x + 5$
(B) $3x - 4 = 2x + 8$
(C) $5(x - 1) = 5x - 5$
(D) $4x + 1 = 4x - 1$

9 Simplify $\sqrt{72}$.

(A) $6\sqrt{2}$
(B) $12\sqrt{6}$
(C) $8\sqrt{9}$
(D) $36\sqrt{2}$

10 If $f(x) = 2x^2 - 3x + 1$, what is $f(-2)$?

(A) $3$
(B) $9$
(C) $15$
(D) $-1$

## Part B — Math I Mastery

11. Which table represents an exponential function?

(A)

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $y$ | 3 | 6 | 9 | 12 |

(B)

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $y$ | 5 | 10 | 20 | 40 |

(C)

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $y$ | 2 | 5 | 10 | 17 |

(D)

| $x$ | 0 | 1 | 2 | 3 |
|---|---|---|---|---|
| $y$ | 9 | 7 | 5 | 3 |

12. The function $g(x) = 120(0.85)^x$ models a phone battery after $x$ hours. Which statement is true?

(A) The initial battery level is 85% and it grows by 120% each hour.
(B) The initial battery level is 120 and it decreases by 15% each hour.
(C) The initial battery level is 120 and it decreases by 85% each hour.
(D) The initial battery level is 0.85 and it grows by 20% each hour.

13. A line is perpendicular to $y = \frac{4}{5}x - 2$ and passes through $(8, 1)$. Which equation represents the line?

(A) $y = -\frac{5}{4}x + 11$
(B) $y = \frac{4}{5}x - \frac{27}{5}$
(C) $y = -\frac{4}{5}x + \frac{37}{5}$
(D) $y = \frac{5}{4}x - 9$

14. Which system has the solution $(2, -1)$?

(A) $\begin{cases} x + y = 3 \\ 2x - y = 5 \end{cases}$

(B) $\begin{cases} x - y = 3 \\ 3x + y = 6 \end{cases}$

(C) $\begin{cases} 2x + y = 3 \\ x + 2y = 1 \end{cases}$

(D) $\begin{cases} x + 2y = 0 \\ 4x - y = 9 \end{cases}$

15. Solve the system: $\begin{cases} 2x + y = 11 \\ x - y = 1 \end{cases}$.

(A) $(3, 5)$
(B) $(4, 3)$
(C) $(5, 1)$
(D) $(2, 7)$

16. A gym charges a sign-up fee plus a monthly fee. The cost is \$95 after 2 months and \$185 after 5 months. What is the monthly fee?

(A) \$20
(B) \$25
(C) \$30
(D) \$45

17. The first term of an arithmetic sequence is $-7$ and the common difference is 5. Which formula gives the $n$th term?

(A) $a_n = -7 + 5n$
(B) $a_n = -7 + 5(n-1)$
(C) $a_n = 5(-7)^{n-1}$
(D) $a_n = -7(5)^{n-1}$

18. The first term of a geometric sequence is 6 and the common ratio is $\frac{1}{3}$. What is the fifth term?

(A) $\frac{2}{9}$
(B) $\frac{2}{27}$
(C) $\frac{6}{27}$
(D) $\frac{1}{162}$

19. Which interval represents $x \geq -2$ and $x < 5$?

(A) $(-2, 5)$
(B) $[-2, 5)$
(C) $(-2, 5]$
(D) $[-2, 5]$

20. The two-way table shows student activity choices.

| | Art | Robotics | Total |
|---|---|---|---|
| Grade 9 | 18 | 27 | 45 |
| Grade 10 | 22 | 13 | 35 |
| Total | 40 | 40 | 80 |

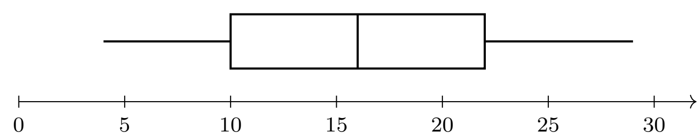
What percentage of Grade 9 students chose Robotics?

(A) 27%
(B) 33.75%
(C) 60%
(D) 67.5%

21 A data set is $\{4, 6, 7, 8, 10, 12, 25\}$. Which statement is true?

(A) The mean is less than the median because 25 is an outlier.
(B) The mean is greater than the median because 25 is an outlier.
(C) The mean equals the median.
(D) The range is 12.

22 The box plot below represents a data set. What is the interquartile range?




(A) 6
(B) 12
(C) 18
(D) 25

23 A scatter plot has a line of best fit $y = 4.8x + 12.5$. Which is the best prediction when $x = 9$?

(A) 43.2
(B) 55.7
(C) 108
(D) 17.3

24 Triangle $ABC$ has vertices $A(1, 2)$, $B(5, 2)$, and $C(1, 7)$. What is the length of $\overline{BC}$?

(A) 3
(B) $\sqrt{41}$
(C) 9
(D) $\sqrt{20}$

25 What transformation maps $(x, y)$ to $(-x, y)$?

(A) Reflection across the $x$-axis
(B) Reflection across the $y$-axis
(C) Rotation 90° clockwise
(D) Translation left $x$ units

26 Which set of side lengths can form a triangle?

(A) $4, 8, 13$
(B) $5, 6, 11$
(C) $7, 9, 12$

(D) $3, 4, 8$

27. If two triangles have two pairs of corresponding sides congruent and the included angle congruent, which criterion proves the triangles congruent?

(A) SSS
(B) SAS
(C) ASA
(D) AAS

28. Which statement best describes the solution region for $y < -2x + 4$?

(A) Solid boundary line, shade above.
(B) Dashed boundary line, shade above.
(C) Solid boundary line, shade below.
(D) Dashed boundary line, shade below.

29. Which expression represents $(f + g)(x)$ when $f(x) = 3x - 8$ and $g(x) = x^2 + 5x + 1$?

(A) $x^2 + 8x - 7$
(B) $x^2 + 2x + 9$
(C) $3x^3 + 15x^2 + 3x - 8$
(D) $x^2 + 5x - 7$

30. The graph of $y = 2^x$ is translated down 5 units. Which equation represents the new graph?

(A) $y = 2^{x-5}$
(B) $y = 2^{x+5}$
(C) $y = 2^x - 5$
(D) $y = 2^x + 5$

## Part C — Enhanced Math I Readiness

31 Simplify $(3\sqrt{5})(2\sqrt{20})$.

(A) 30
(B) 60
(C) $30\sqrt{5}$
(D) $6\sqrt{25}$

32 A right triangle has legs 9 and 12. What is the length of the hypotenuse?

(A) 15
(B) $\sqrt{63}$
(C) 21
(D) $3\sqrt{7}$

33 A ladder is 13 feet long and reaches a window 12 feet above the ground. How far is the bottom of the ladder from the wall?

(A) 1 ft
(B) 5 ft
(C) 7 ft
(D) 25 ft

34 Solve: $|2x - 7| = 11$.

(A) $x = 9$ only
(B) $x = -2$ only
(C) $x = 9$ or $x = -2$
(D) No solution

35 Which inequality represents $|x - 3| \leq 8$?

(A) $-5 \leq x \leq 11$
(B) $x \leq -5$ or $x \geq 11$
(C) $-8 \leq x \leq 8$
(D) $x \leq -11$ or $x \geq 5$

36 Solve the system: $\begin{cases} 3x - 2y = 6 \\ 5x + y = 23 \end{cases}$.

(A) $(4, 3)$
(B) $(2, 13)$
(C) $(5, -2)$
(D) $(6, -7)$

37 Which point is in the solution set of $\begin{cases} y \geq x - 1 \\ y < -2x + 5 \end{cases}$ ?

(A) $(0, 0)$
(B) $(3, 5)$
(C) $(4, -1)$
(D) $(2, 1)$

38 If $f(x) = 2x + 1$ and $g(x) = x^2 - 4$, what is $(fg)(3)$?

(A) 17
(B) 35
(C) 49
(D) 63

39 A geometric sequence has $a_2 = 18$ and $a_5 = 486$. What is the common ratio?

(A) 2
(B) 3
(C) 6
(D) 9

40 Which function has an average rate of change of 7 on the interval $[1, 4]$?

(A) $f(x) = 2x + 5$
(B) $f(x) = x^2$
(C) $f(x) = 3^x$
(D) $f(x) = 7x - 4$

41 The residual for a data point is:

(A) actual $y$ minus predicted $y$
(B) predicted $x$ minus actual $x$
(C) slope divided by intercept
(D) the mean of the $x$-values

42. A line of best fit predicts 84 when the actual value is 79. What is the residual?

(A) 5
(B) $-5$
(C) 163
(D) 0.94

43. Which transformation maps $(x, y)$ to $(y, -x)$?

(A) Rotation 90° clockwise about the origin
(B) Rotation 90° counterclockwise about the origin
(C) Reflection across $y = x$
(D) Reflection across the $x$-axis

44. Quadrilateral $PQRS$ has vertices $P(0, 0)$, $Q(4, 1)$, $R(5, 5)$, and $S(1, 4)$. Which is the strongest classification?

(A) Trapezoid
(B) Rectangle
(C) Rhombus
(D) Square

45. In a proof, which additional fact would prove $\triangle ABC \cong \triangle DCB$ by SAS if $AB \cong DC$ and $\angle ABC \cong \angle DCB$?

(A) $AC \cong DB$
(B) $BC \cong CB$
(C) $\angle BAC \cong \angle CDB$
(D) $AB \parallel DC$

46. For which value of $k$ will the system below have no solution?

$$\begin{cases} y = 3x + 2 \\ ky - 6x = 8 \end{cases}$$

(A) $k = 1$
(B) $k = 2$
(C) $k = 3$
(D) $k = 6$

47 The function $h(x) = |x - 2| + 3$ is graphed in the coordinate plane. Which statement is true?

(A) The range is $[3, \infty)$.
(B) The domain is $[3, \infty)$.
(C) The vertex is $(-2, 3)$.
(D) The function is linear.

48 What is the distance between $A(-6, 2)$ and $B(2, 8)$?

(A) $\sqrt{28}$
(B) $\sqrt{52}$
(C) 10
(D) 14

49 Which point represents an exact intersection of $f(x) = 2^x - 1$ and $g(x) = x + 1$?

(A) $(0, 1)$
(B) $(1, 2)$
(C) $(2, 3)$
(D) $(3, 4)$

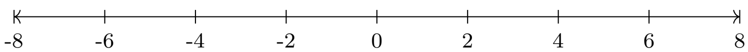
50 Triangle $PQR$ has side lengths $PQ = 5$, $QR = 12$, and $PR = 13$. Which statement must be true?

(A) $\triangle PQR$ is acute.
(B) $\triangle PQR$ is a right triangle.
(C) $\triangle PQR$ is obtuse.
(D) The side lengths cannot form a triangle.

## Graphing Questions

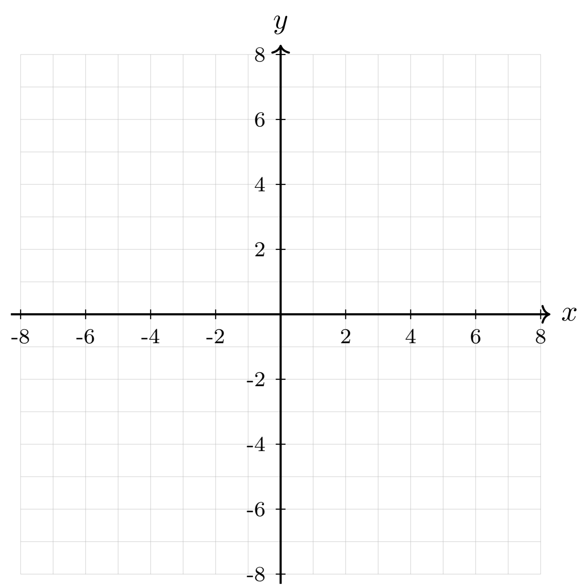
### Constructed Response — Graphing and Coordinate Reasoning

1. Graph the solution to $-3(x-2) \leq 15$ on a number line. Then write the solution in interval notation.



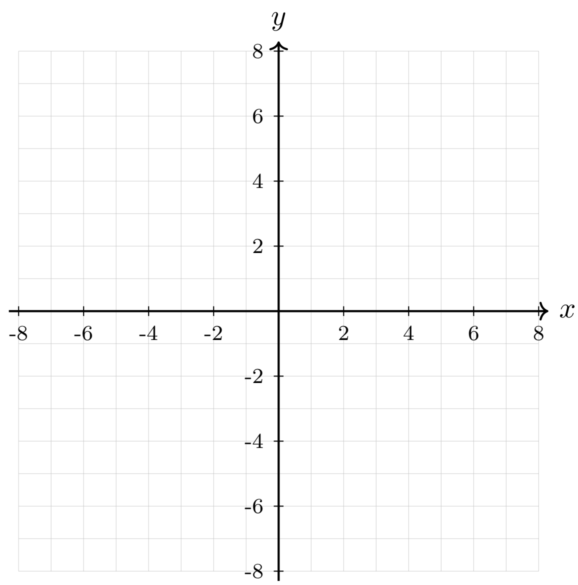

Interval notation: ______________________________

2. Graph the inequality $y \geq -\frac{1}{2}x + 3$.

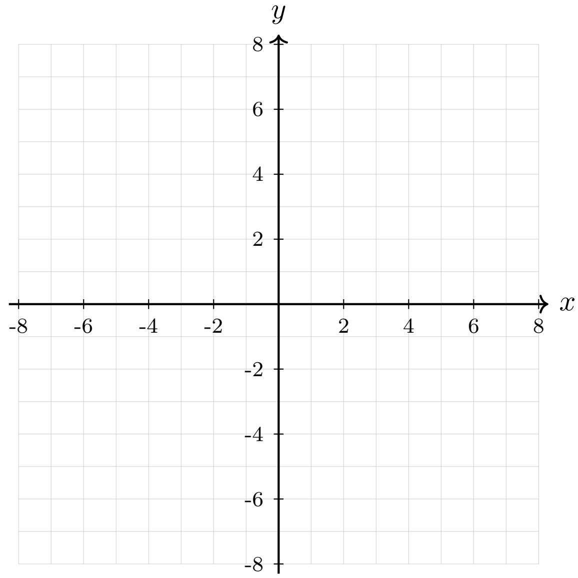


3. Graph the system of inequalities.

$$\begin{cases} y < 2x - 1 \\ y \geq -x + 4 \end{cases}$$



4 Plot the points $A(-4, 1)$, $B(2, 1)$, and $C(-1, 5)$. Then rotate the triangle $90°$ counterclockwise about the origin and label the image $A'B'C'$.



## Free Response Questions

### Free Response — Modeling, Reasoning, and Enhanced Challenge

1 **Equation Reasoning.** Solve the equation and determine whether it has one solution, no solution, or infinitely many solutions.

$$4(3x - 5) + 7 = 2(6x - 9) - 1$$

Show each step and explain your conclusion.

2 **Linear Modeling.** A tutoring center charges an enrollment fee plus a constant hourly rate. A student pays \$150 for 4 hours and \$255 for 9 hours.

(a) Write a linear equation for the total cost $C$ after $h$ hours.

(b) Interpret the slope and $C$-intercept in context.

(c) Predict the cost for 14 hours.

3 **Systems and Context.** A school sells adult tickets and student tickets for a performance. Adult tickets cost \$12 and student tickets cost \$8. The school sells 210 tickets and collects \$2,120.

(a) Define variables and write a system of equations.

(b) Solve the system.

(c) Explain what the solution means in context.

4 **Linear vs. Exponential.** Two savings plans are shown below.

$$\text{Plan A: } A(n) = 250 + 40n \qquad \text{Plan B: } B(n) = 250(1.12)^n$$

(a) Identify which plan is linear and which is exponential.

(b) Find the amount in each plan after 5 years.

(c) Explain which plan is growing faster at year 5 and why.

5 **Sequences.** A geometric sequence begins $5, 15, 45, 135, \ldots$

(a) Write an explicit formula for the $n$th term.

(b) Find the 10th term.

(c) Compare this sequence to the arithmetic sequence $a_n = 5 + 10(n-1)$ at $n = 10$.

6 **Statistics and Outliers.** The data set gives quiz scores:

$$62,\ 70,\ 74,\ 76,\ 79,\ 80,\ 82,\ 84,\ 86,\ 88,\ 91,\ 100$$

(a) Find the mean and median.

(b) Find the interquartile range.

(c) Explain whether the mean or median better represents the typical score.

7 **Bivariate Data.** A line of best fit for a data set is $y = -3.6x + 92$.

(a) Interpret the slope in context if $x$ is hours of screen time and $y$ is test score.

(b) Predict the score for $x = 6$.

(c) If the actual score is 68, find and interpret the residual.

8 **Pythagorean Theorem and Radicals.** A rectangular park is 48 meters long and 20 meters wide.

(a) Find the diagonal distance across the park.

(b) A student walks from one corner to the opposite corner and then back along the outside edges. How many meters shorter is the diagonal path than the edge path from one corner to the opposite corner?

(c) Explain why the Pythagorean Theorem applies.

9 **Coordinate Geometry Classification.** Quadrilateral $ABCD$ has vertices

$$A(-2,1), \quad B(3,3), \quad C(5,-2), \quad D(0,-4).$$

Classify the quadrilateral as specifically as possible. Support your answer using slope, distance, or both.

10 **Transformation and Congruence Proof.** Triangle $ABC$ has vertices

$$A(-3,2), \quad B(-1,6), \quad C(2,3).$$

It is reflected across the $y$-axis and then translated 4 units down.

(a) Give the coordinates of $A'B'C'$.

(b) Explain why $\triangle ABC \cong \triangle A'B'C'$.

(c) Name the rigid motions used.

11 **Enhanced Challenge: Nonlinear Systems Graphically.** Consider the functions

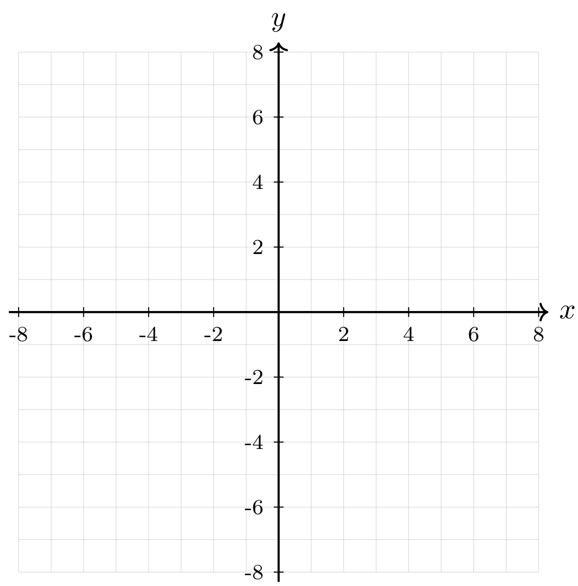
$$f(x) = 2^x - 1, \qquad g(x) = x + 1.$$

(a) Complete a table for $x = -1, 0, 1, 2, 3$ for both functions.

(b) Estimate all intersection points from the table or a graph.

(c) Explain what each intersection means as a solution to a system.

| $x$ | $-1$ | $0$ | $1$ | $2$ | $3$ |
|---|---|---|---|---|---|
| $f(x)$ | | | | | |
| $g(x)$ | | | | | |



## Teacher Scoring Guide

**Suggested point values:** Multiple Choice 50 points; Graphing 12 points; Free Response 44 points. Total: 106 points.

**IUSD alignment used for this diagnostic:**

- **Math I:** functions, linear and exponential functions, arithmetic and geometric sequences, equations and inequalities, systems, one- and two-variable statistics, constructions and transformations, and triangle congruence.
- **Enhanced Math I:** all Math I expectations, plus stronger fluency with systems, radicals, Pythagorean Theorem applications, bivariate data, transformations, coordinate geometry, and geometric proof.

**Two-Gate Placement Protocol**

| Decision Gate | Use These Items | Decision Rule |
|---|---|---|
| Gate 1: Math I qualification | MC 1-30, Graphing 1-2, FR 1-3. Maximum 48 points. | 36-48: qualified for Math I. 30-35: borderline; begin Math I with targeted support. Below 30: not yet ready for Math I without remediation. |
| Gate 2: Enhanced Math I qualification | Only apply if Gate 1 is passed. Use MC 31-50, Graphing 3-4, FR 4-11. Maximum 58 points. | 44-58 and MC Part C at least 15/20: qualified for Enhanced Math I. 37-43: borderline Enhanced; targeted preparation recommended. Below 37: place in Math I or build first. |

**Important placement note:** A high total score should not override Gate 1. If a student is weak on linear equations, functions, rates of change, and systems, the student is not ready for Enhanced Math I even if some later topics are familiar.

## Study Plan by Diagnostic Band

**Lesson length assumption:** one lesson is 90 minutes. If lessons are 60 minutes, multiply the lesson count by about 1.5.

| **Band** | **Recommended Lessons** | **Focus** |
|---|---|---|
| Below Math I readiness | 20-28 lessons | Integer/rational operations, equation solving, inequalities, graphing lines, slope, function notation, basic word problems, and academic vocabulary. Goal: pass Gate 1 reliably. |
| Borderline Math I | 12-16 lessons | Linear equations, slope and intercepts, function vs. non-function, proportional relationships, basic systems, and interpreting tables/graphs. Goal: enter Math I with support rather than struggle from the first unit. |
| Math I ready, not Enhanced ready | 16-24 lessons | Systems of equations and inequalities, linear vs. exponential modeling, arithmetic/geometric sequences, statistics, scatter plots, transformations, radicals, and Pythagorean applications. Goal: build Gate 2 readiness. |
| Borderline Enhanced | 8-12 lessons | Enhanced-level problem solving: multi-step systems, parameter reasoning, interval notation, absolute value, residuals, coordinate geometry classification, proof reasoning, and nonlinear systems graphically. Goal: reduce careless errors and strengthen explanations. |
| Enhanced ready | 4-6 lessons | Maintenance and enrichment: challenge modeling, proof writing, mixed-topic review, and test-taking accuracy. Goal: keep the student sharp and prepare for the first Enhanced units. |

## Module Map

| Module | Lessons | When to assign |
|---|---|---|
| Algebra foundation | 4-6 | Misses MC 1-4, 8, FR1; weak equation steps or sign errors. |
| Functions and linear models | 4-6 | Misses MC 5-7, 11-16, FR2; weak slope, intercept, function notation, or context interpretation. |
| Systems and inequalities | 3-5 | Misses MC 14-15, 28, 36-37, 46, Graphing 2-3, FR3. |
| Exponential functions and sequences | 3-5 | Misses MC 11-12, 17-18, 30, 39-40, FR4-5. |
| Statistics and bivariate data | 3-5 | Misses MC 20-23, 41-42, FR6-7. |
| Radicals and Pythagorean Theorem | 3-4 | Misses MC 9, 31-33, FR8; this is a key Enhanced Math I signal. |
| Transformations, coordinate geometry, proof | 5-7 | Misses MC 24-27, 43-45, Graphing 4, FR9-10; weak reasoning or unsupported claims. |
| Enhanced challenge and mixed review | 2-4 | Gate 2 score is close to the cutoff; use for nonlinear systems, function transformations, mixed modeling, and explanation quality. |

# Answer Key and Selected Free Response Answers

Use this page for quick grading. The selected free-response answers show the essential result; full credit should still depend on clear work and reasoning.

### Multiple Choice Answer Key

| Q | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans | A | A | C | B | B | B | A | C | A | C |

| Q | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans | B | B | A | D | B | C | B | B | B | C |

| Q | 21 | 22 | 23 | 24 | 25 | 26 | 27 | 28 | 29 | 30 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans | B | B | B | B | B | C | B | D | A | C |

| Q | 31 | 32 | 33 | 34 | 35 | 36 | 37 | 38 | 39 | 40 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans | B | A | B | C | A | A | A | B | B | D |

| Q | 41 | 42 | 43 | 44 | 45 | 46 | 47 | 48 | 49 | 50 |
|---|---|---|---|---|---|---|---|---|---|---|
| Ans | A | B | A | C | B | B | A | C | C | B |

### Selected Free Response Answers

- FR1: No solution, because simplifying gives $12x - 13 = 12x - 19$.
- FR2: $C = 21h + 66$; slope is hourly rate \$21, intercept is enrollment fee \$66; $C(14) = 360$.
- FR3: $a + s = 210$, $12a + 8s = 2120$; adult 110, student 100.
- FR4: Plan A linear, Plan B exponential; $A(5) = 450$, $B(5) \approx 440.59$.
- FR5: $a_n = 5(3)^{n-1}$; $a_{10} = 98,415$; arithmetic value is 95.
- FR6: Mean 81, median 81; IQR 12; either measure is reasonable because the set is fairly balanced.
- FR7: Slope means each extra hour predicts a 3.6 point lower score; predicted 70.4; residual $-2.4$.
- FR8: Diagonal 52 m; edge path 68 m, so diagonal is 16 m shorter.
- FR9: Square; all four side lengths are $\sqrt{29}$ and adjacent slopes are negative reciprocals.
- FR10: $A'(3, -2)$, $B'(1, 2)$, $C'(-2, -1)$; reflection and translation preserve distance and angle measure.
- FR11: Intersections at approximately $(-1.7, -0.7)$ and $(2, 3)$.